ETL

Business Requirements Analysis for a Retail Analytics Platform

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1) Business Problem Analysis

The company needs a centralized, visual dashboard because managers currently have to manually dig through spreadsheets to understand daily sales performance across stores, which slows decision-making and can cause missed opportunities, such as failing to catch a decline in a product category in time to launch a corrective promotion. This matters to the organization because faster, clearer visibility into sales protects revenue, saves management time, and builds a foundation for a more data-driven culture as the business grows. The main stakeholders who would benefit are regional and store managers monitoring their own performance, the marketing team adjusting campaigns based on target attainment, and senior leadership looking for strategic oversight. With this information readily available, managers could decide when to launch a promotion for an underperforming product, reallocate marketing budget toward struggling stores or regions, adjust inventory and restocking priorities, and revise sales targets based on historical trend comparisons.

Some questions we'd ask right after:

- What is the budget and timeline for this project?
- What database holds the sales data? What system holds promotions data, and in what format?
- What counts as "underperforming"? Is there a threshold or should the system flag anomalies automatically?
- What does "mobile access" mean? A responsive web dashboard, or a native app?
- What is the security/access requirements (should a regional manager only see their own region's data)?
- Which report formats does marketing need?
- Which colors you want to be used in the dashboard?

2) Business Objectives

Business Objective	Why is important
Improve the effectiveness of marketing campaigns based on store performance.	Marketing campaigns can be adjusted to maximize their impact and improve sales performance across stores.
Monitor store performance against sales targets to support timely business decisions.	Managers can quickly identify stores that need additional support or strategic actions.
Optimize commercial decision-making through the early detection of underperforming products.	Detecting declining products early helps prevent revenue loss through timely promotions or inventory adjustments.
Compare sales performance across different regions to support strategic planning.	Regional comparisons help management identify high-performing areas and allocate resources more effectively.
Measure the effectiveness of pricing and discount strategies on sales performance.	Have a better adjustment on campaigns according of how well they are performing due to their pricing and discount percentages.
Provide an intuitive and user-friendly dashboard for managers with different levels of technical experience.	A simple, easy-to-use dashboard allows managers to access information quickly and facilitates more agile decision-making.

3) Analytical Questions

Business question	Information required
Which product categories are performing well, and which ones are struggling?	Sales by product category, with trend over time
Which pricing and discount strategies are generating the highest sales lift?	Pricing history, discount rates, and sales data before and during promotional periods
Which specific stores and regions are underperforming?	Sales by region and by individual store
Are promotions actually improving sales?	Promotion's data joined with sales data
Which stores are hitting the targets, and which aren't?	Actual sales vs. target, by store

4) Functional Requirements

- Display total sales KPI with selectable time ranges.
- Show sales broken down by product category, highlighting underperforming categories.
- Allow filtering by region and by individual store.
- Provide trend charts with comparison to previous month or year.
- Show target vs. actual performance by store for marketing use.
- Allow drill-down from region, store and product level of detail.
- Allow users to export reports.
- Display a clean navigation with minimal menus.

5) Non-functional Requirements

- **Performance:** Dashboard views and filters should load within a few seconds, even with large historical datasets.
- **Security:** Role-based access control so managers only see data relevant to their region or store.
- **Availability:** The system should be available 24/7.
- **Scalability:** The architecture should support adding more stores, regions, or data sources without a major redesign.
- **Reliability:** Data pipelines must ensure accuracy and consistency between the sales and promotions sources.
- **Mobile Compatibility:** The dashboard should render well, remain usable and available on phones.

6) Data Requirements

Information Needed	Data Required	Possible Source	Why is it Needed?
Sales Performance	Sales transactions (store, product, date, amount)	Point of Sale System	To calculate sales performance, compare actual sales with sales targets, and monitor store and regional performance.
Marketing Campaign Data	Campaign ID, campaign dates, promoted products, campaign type	Marketing Campaign Management System	To evaluate the effectiveness of marketing campaigns and measure their impact on sales performance.
Product Performance	Product catalog, product category, sales by product	Inventory Database + POS System	To identify high- and low-performing product categories and support commercial decisions.
Pricing and Discount Data	Pricing history, discount rates and promotion periods by product	Promotions Database Product Prices Database	To evaluate the impact of pricing changes and discounts on sales performance, and identify which strategies generate the highest revenue raise
Sales Targets	Sales targets by store and period	Sales Planning / Business Management System	To compare actual sales with expected targets and measure store performance.
Regional Information	Store ID, region, store location	Store Master Data	To compare sales performance across different regions and support strategic planning.

Dashboard Usability	User satisfaction survey results, usability test results	User Feedback / Survey Platform	To evaluate whether managers can easily use the dashboard without additional training.

A. Do all the required data come from a single source? Explain:

No. Each type of information comes from a different system, monthly sales comes from the Point of Sale system, product performance requires combining the product catalog with sales data from the Inventory Database, and customer behavior comes from the CRM. Sales data lives in one system while promotions data lives in another.

B. What challenge does the organization face if the required information is distributed across multiple systems?

The main challenge is data integration. The organization needs to combine data that lives in different systems and formats. This introduces risks such as inconsistent or duplicate data, mismatched refresh schedules, extra ETL/pipeline development effort, higher maintenance cost, and the possibility of delays or errors if the integration isn't well designed. Without addressing this, the dashboard risks showing incomplete or inconsistent numbers.

C. Can a dashboard answer these questions without integrating multiple data sources? Explain.

Not effectively. A dashboard connected to only one source could show sales totals, but it couldn't tell you why a product is underperforming or who is buying it and how that's changing. To answer cross-cutting questions the data first needs to be integrated into a common model.

7) User Stories

As a marketing team member, I want to know which products are the ones with the least sales, so that I can know which products I should promote to raise their sales.

As a regional manager, I want to see a summary of all my stores performance as soon as I log in, so that I can immediately identify if any store requires my immediate attention to raise their performances.

As a regional manager, I want to be able to know the sales of the products over time, so that I can compare the sales of the products across different time ranges and realize which ones are having a decline over time.

As a marketing team member, I want to download a report of which stores are hitting their sales targets and which aren't, so that I can share it with my team and adjust our campaigns accordingly.

As a regional manager, I want to filter sales data by region and by store, so that I can focus the analysis on the specific areas in which I am responsible for and identify which ones need more attention.

User	Decision to make	Information needed
Marketing team member	Which products to run a promotion	Total sales from each individual product
Regional manager	Focus on the stores with the worst performance to take action to raise their performance	Stores performances
As a regional manager	Have into consideration the products	Historical sales
Marketing team member	Adjust the campaigns according to the store's performances	Stores performances Targets by store
As a regional manager	Which specific region or store requires immediate attention	Sales data per store Sales data per region

8) KPIs

Business Objective	KPIs
Improve the effectiveness of marketing campaigns by aligning them with store performance.	Marketing Campaign Performance. **(Sales during campaign − Sales before campaign) ÷ Sales before campaign × 100**
Monitor store performance against sales targets to support timely business decisions.	Store Target Achievement. Daily Store Performance. **Actual Sales ÷ Sales Target × 100** **Daily Sales ÷ Daily Sales Target ×100**
Optimize commercial decision-making through the early detection of underperforming products.	Product Category Performance. **(Current Category Sales − Previous Category Sales) ÷ Previous Category Sales ×100**
Compare sales performance across different regions to support strategic planning.	Regional sales performance. **Regional Sales ÷ Total Sales × 100**
Measure the effectiveness of pricing and discount strategies on sales performance	Promotional Sales Lift **(Sales during discount period − Sales before discount) ÷ Sales before discount × 100**
Provide an intuitive and user-friendly dashboard for managers with different levels of technical experience.	Dashboard usability. **Successful Tasks Completed ÷ Total Assigned Tasks × 100.**

- **Which KPIs require data from multiple systems?**

The KPIs that require data from multiple systems are those that need information from different business areas to provide a complete view of performance.

Marketing Campaign Performance; requires data from the sales system and the marketing campaign management platform.

Store Target Achievement and Daily Store Performance, require sales data from the stores and sales targets defined by the business.

Inventory Availability requires data from the inventory system and sales information to verify whether product availability meets customer demand.

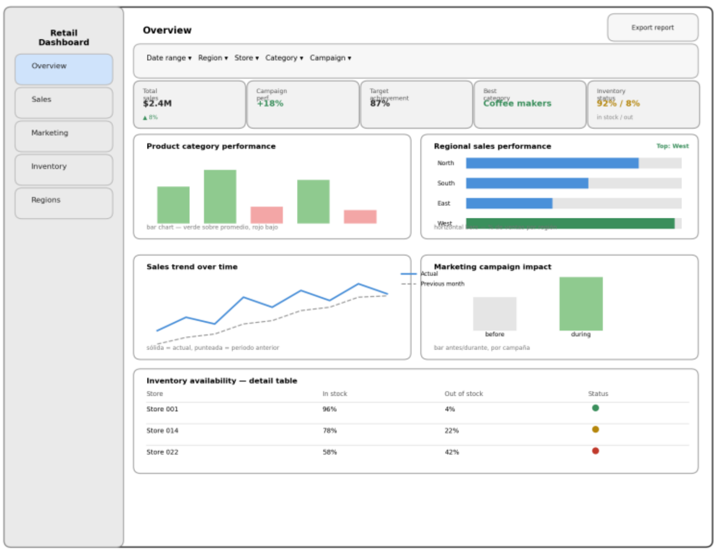
The remaining KPIs can generally be calculated using data from a single source or closely related datasets.

- **Which KPI would be the most difficult to calculate? Why?**

The most difficult KPI to calculate would be Marketing Campaign Performance because it requires combining data from two different sources: the Promotions System and the Point of Sale (POS) System.

To measure this KPI, sales before and during each campaign must be compared to determine whether the promotion had a positive impact. It is also important to match the data by product, store, and time period to ensure an accurate comparison.

9) Dashboard Wireframe



The wireframe follows a manager-first hierarchy: the sidebar organizes information by business domain (Sales, Marketing, Inventory, Regions) rather than by chart type, so managers navigate by the question they have, not by the visualization. Global filters (date range, region, store, category, campaign) sit at the top and affect the entire view at once, avoiding the need to filter each chart individually.

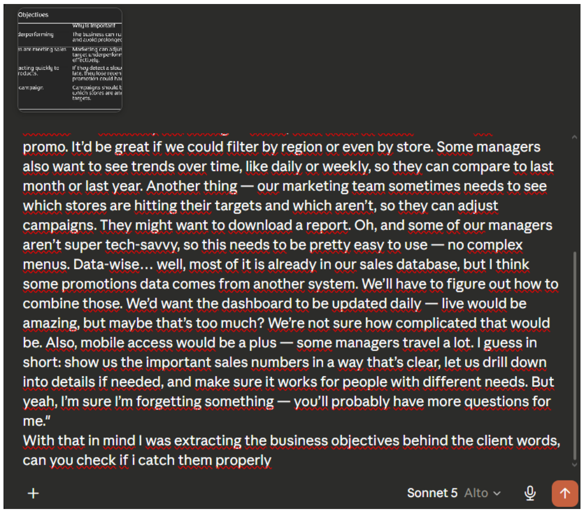
The top KPI row shows the five business-facing metrics defined in Activity 6. Dashboard Usability was intentionally excluded from this view, since it is an internal product metric relevant to the development team, not to store or marketing decisions.

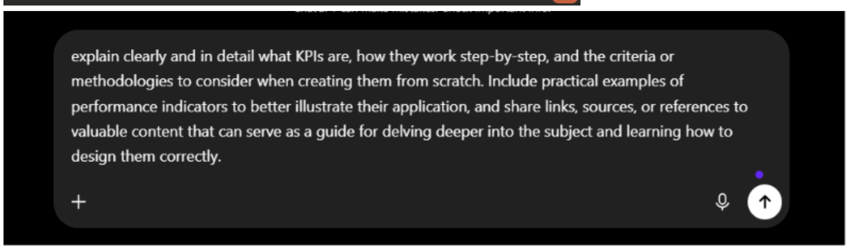
Charts were chosen to match the nature of each question: bar charts with color thresholds for category and campaign performance (quickly flags underperformance), horizontal bars for regional comparison, a line chart with a previous-period reference for trend analysis, and a color-coded table for inventory, since store-level detail is better read as rows than as a chart.

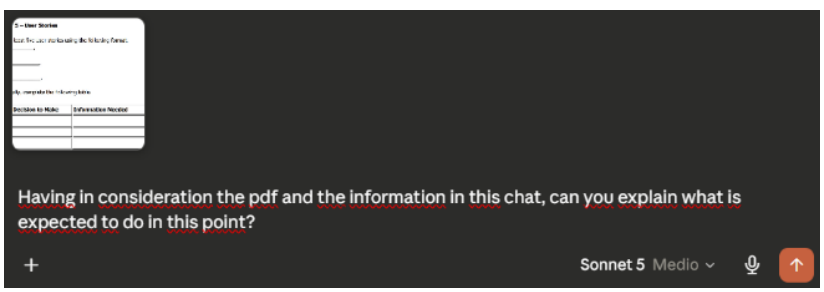
10) Requirements-to-Data Mapping Table

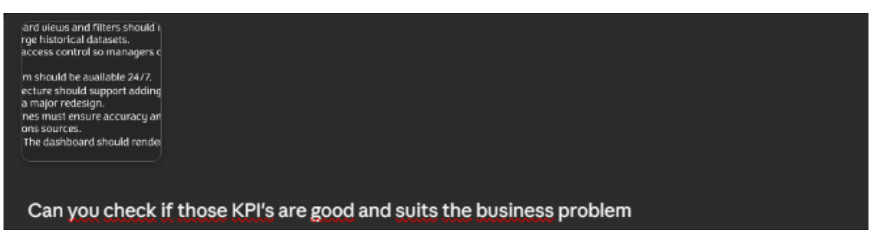
Business Requirement	**Data Source**	Transformation Needed	Expected Output
Track monthly sales performance to compare against targets and previous periods	POS System	Aggregate transaction-level sales by store, product, and date; standardize date formats; calculate monthly totals	Monthly sales summary table.
Detect underperforming product categories early to trigger promotions	Inventory Database	Join product catalog with sales on product ID, aggregate sales by category and time period.	Product category performance table.
Compare sales performance across stores/regions to support strategic planning	POS System and store master data	Join sales data with the store dimension table to append region; aggregate sales by store/region; calculate target achievement.	Store/region performance table.
Evaluate whether promotions are improving sales	Promotions System and POS	Join promotions data with sales; compare sales before vs. during promotion.	Promotion effectiveness table.
Measure the impact of pricing and discount strategies on sales performance	Promotions System and POS	Join pricing/discount records with sales data by product and time period; compare sales before vs. during discount periods; compute revenue lift per strategy	Pricing and discount effectiveness report

11) AI prompts









Actúa como un Diseñador Senior de UX/UI y Analista de Inteligencia de Negocios (BI). Necesito que diseñes la estructura de un "Dashboard Wireframe" de baja fidelidad (Low-Fidelity Layout / Wireframe) dirigido a gerentes y toma de decisiones. A continuación te comparto los Objetivos de Negocio y sus KPIs asociados para este dashboard:

1. Marketing Campaign Performance:
    - Objetivo: Improve the effectiveness of marketing campaigns by aligning them with store performance.
    - Fórmula KPI: (Sales during campaign − Sales before campaign) ÷ Sales before campaign × 100
2. Store Target Achievement:
    - Objetivo: Track whether stores meet their sales targets.
    - Fórmula KPI: Actual Sales ÷ Sales Target × 100
3. Product Category Performance:
    - Objetivo: Optimize commercial decision-making through the early detection of underperforming products.
    - Fórmula KPI: Category Sales ÷ Total Sales × 100
4. Regional Sales Performance:
    - Objetivo: Compare sales performance across different regions to support strategic planning.
    - Fórmula KPI: Regional Sales ÷ Total Sales × 100
5. Inventory Availability:
    - Objetivo: Improve inventory visibility to reduce stock shortages and overstock situations.
    - Fórmula KPI: Out-of-stock % / In-stock % (Visualización de stock/inventario)
6. Dashboard Usability (Métrica interna/operativa):
    - Objetivo: Provide an intuitive and user-friendly dashboard for managers with different levels of technical experience.
    - KPI: Successful Tasks Completed ÷ Total Assigned Tasks × 100

REQUERIMIENTOS DE LA ENTREGABLE:
Por favor, genera la propuesta de diseño estructurada en las siguientes secciones:

1. Navigation Layout & Layout Structure:
    - Describe la navegación global (Navbar superior, Sidebar lateral, etc.).
    - Define la distribución por paneles/secciones para organizar la información de forma lógica.
2. Global Filters:
    - Lista los filtros que deben estar disponibles arriba para que los gerentes segmenten la información (ej. Fecha/Rango, Región, Tienda, Categoría de producto, Campaña).
3. Main KPIs Cards (Sección Superior):
    - Especifica las tarjetas de KPIs principales (Scorecards) mostrando qué valor, tendencia o comparativo se debe ver en cada una.
4. Recommended Charts & Visuals (Sección Central/Inferior):
    - Para cada objetivo/KPI, sugiere el tipo de gráfico ideal (ej. Bar Chart, Line Chart, Heatmap, Table con indicadores de color, etc.) y qué ejes o dimensiones debe llevar.
5. Low-Fidelity ASCII Wireframe:
    - Dibuja un esquema visual claro en formato de bloque de código (ASCII/Text Wireframe) representando la pantalla del Dashboard con sus tarjetas, filtros, sidebar y gráficos integrados.

No se requiere código de programación ni implementación técnico-frontend, solo el diseño conceptual y la arquitectura de la interfaz.

Y como seguimiento, para pedir la versión visual en imagen usaste:

No, dame una imagen chévere como ahora, la imagen claramente no es funcional, pero sí que se vean cómo quedaría.

12) After completing this activity, do you think building a dashboard is the first step in a Data Engineering project? Explain your answer.

Definitely not, during this exercise, we learned that the client's first request ("we want a dashboard") was only a general idea of what they really needed. Before designing the dashboard, we had to understand the business problem, define the business objectives, identify the analytical questions, review the functional and non-functional requirements, understand the data sources, and define the KPIs. Only after doing all of that could, we create a useful dashboard wireframe.

We also learned that a dashboard can show a lot of information, but not all of it is useful for the client. If we had started building the dashboard, we probably would have included metrics that looked useful from a technical point of view but were not important for managers. For example, we decided not to include the Dashboard Usability KPI because it is useful for the development team, not for managers making business decisions. This showed us how important it is to understand who will use the dashboard and what information they really need.

In conclusion, building a dashboard is not the first step in a Data Engineering project. It is one of the last steps. First, we need to understand the business problem, the stakeholders, the available data, and the project requirements. Only then can we build a dashboard that helps users make better decisions instead of simply showing a lot of data.